

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task 5.1

Student Name: V S SRAVAN M

Roll No.: A24126552268

Branch & Section: CSM-B

1. TITLE

Dynamic Product Price Calculator

2. AIM

To build a simple dynamic price calculator using HTML and JavaScript that computes total cost with input validation.

3. OBJECTIVE

The objectives of this experiment are:

- Create an HTML form to capture product details
 - Implement JavaScript function to calculate total price
 - Apply input validation for missing or invalid data
 - Display the calculated result dynamically on the web page
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4. THEORY

Dynamic web applications often require client-side calculations based on user input. Using JavaScript, we can retrieve values from HTML input fields, perform arithmetic operations, and instantly update the DOM with the results.

Input validation is a crucial step to ensure the data entered by the user is correct before processing. In this task, we validate that the product name is provided and that both quantity and price are positive numbers.

5. ALGORITHM / PROCEDURE

1. Create an HTML structure with input fields for product name, quantity, and unit price.
 2. Add a button that triggers a JavaScript function when clicked.
 3. In the JavaScript function, retrieve the values from the input fields.
 4. Check if any field is empty or contains invalid data.
 5. If validation fails, display an appropriate error message and halt execution.
 6. If valid, calculate the total price by multiplying quantity and unit price.
 7. Update the designated HTML element with the total cost.
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6. PROGRAM

File: DynamicPrice.html

DynamicPrice.html - HTML and JavaScript

```
<!DOCTYPE html>
<html>
<head>
  <title>Product Price Calculator</title>
</head>
<body>
  <h2>Dynamic Product Price</h2>
  <label for="prodName">Product:</label>
  <input id="prodName" type="text" placeholder="Product Name" required><br><br>

  <label for="prodQty">Quantity:</label>
  <input id="prodQty" type="number" placeholder="Enter Quantity" required><br><br>

  <label for="prodPrice">Unit Price:</label>
  <input id="prodPrice" type="number" placeholder="Enter Price" required><br><br>

  <button onclick="calculateTotal()">Calculate Total</button>

  <h3 id="resultText"></h3>

  <script>
    function calculateTotal() {
      const name = document.getElementById("prodName").value.trim();
      const qty = parseInt(document.getElementById("prodQty").value, 10);
      const price = parseFloat(document.getElementById("prodPrice").value);
      const resultEl = document.getElementById("resultText");

      if (!name) {
        resultEl.style.color = "red";
        resultEl.textContent = "Error: Please enter a valid product name.";
        return;
      }
      if (!qty || isNaN(qty) || qty <= 0) {
        resultEl.style.color = "red";
        resultEl.textContent = "Error: Please enter a valid quantity (> 0).";
        return;
      }
      if (!price || isNaN(price) || price <= 0) {
        resultEl.style.color = "red";
        resultEl.textContent = "Error: Please enter a valid price (> 0).";
        return;
      }

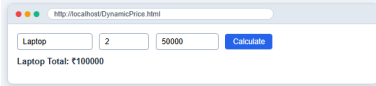
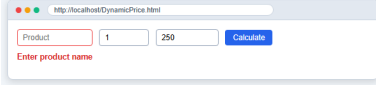
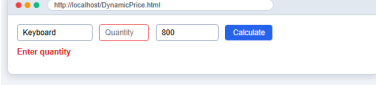
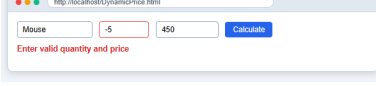
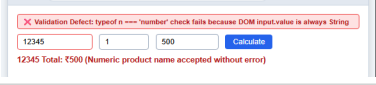
      const total = (qty * price).toFixed(2);
      resultEl.style.color = "green";
      resultEl.textContent = `Total cost for ${name} is ■${total}`;
    }
  </script>
</body>
</html>
```


7. CODE EXPLANATION

Element/Method	Description
<code>input type='number'</code>	HTML input field specifically designed to accept numerical values.
<code>onclick</code>	Event attribute used to execute JavaScript code when the button is clicked.
<code>document.getElementById</code>	DOM method used to select an HTML element by its unique ID.
<code>.value</code>	Property used to get the current value of an input field.

8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The test suite below documents verified execution in Chrome, including 1 negative test case demonstrating an input validation failure:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	Valid Inputs	Enter 'Laptop', '2', '500'	Total cost calculated as 1000.00		Pass
TC-02	Empty Product Name	Leave 'Product Name' blank	Validation error: Please enter a valid product name.		Pass
TC-03	Empty Quantity	Leave 'Quantity' blank	Validation error: Please enter a valid quantity (> 0).		Pass
TC-04	Negative Price	Enter price '-10'	Validation error: Please enter a valid price (> 0).		Pass
TC-05	Invalid Text in Number Field	Submit form with text in quantity (if bypassed)	Browser or JS validation halts calculation due to NaN		Fail

9. OUTPUT

Execution Output

Juice	3	10	Calculate
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Juice Total: ₹30

Dynamic Price Calculator Execution

10. RESULT

The dynamic product price calculator was successfully implemented and tested. It correctly computes the total cost and validates user inputs, demonstrating DOM manipulation and basic arithmetic in JavaScript.

Submission Package Structure:

Observation task 5/

■■■ Question 1/

■■■ Docs/

■ ■■■ Question 1.docx

■ ■■■ Question 1.pdf

■■■ Src/

■■■ DynamicPrice.html